

Long-run growth and welfare effects of public policies with distortionary taxation[☆]

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Abstract

In a one-sector growth model with infrastructure investment, we study the impact of fiscal policies on growth. The government collects taxes on labor income and profits. The government uses these revenues to purchase infrastructure investment, provide utility enhancing government services, and to provide transfer payments to the households. We show that the balanced growth rate is an increasing function of the percent of government revenues that goes to infrastructure. We find that the growth maximizing mix of taxes depends on the elasticity of substitution between inputs. In particular, with distortionary taxes, the growth maximizing tax rate on capital is higher the lower the elasticity of substitution between private physical capital and public capital. In addition, the growth maximizing size of the government is higher when the elasticity of substitution between public and private inputs is low. We also investigate welfare effects of other public expenditures as well. Depending on the elasticity of substitution in production, the

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welfare implications of different public policies can be substantial. © 2001 Elsevier Science B.V. All rights reserved.

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1. Introduction

Economists have long believed there is a connection between fiscal policies and long-run growth. This nexus is still much debated. One point of view suggests that higher taxation (bigger government) unambiguously lowers growth rates and reduces welfare. In this view, raising revenues through taxation distorts private decisions to work and save. Another point of view argues that certain activities of the government may enhance growth or welfare. Public investment in roads, highways, airports, and public expenditures that secure property rights may increase private returns, thus stimulating investment and growth. In addition to the growth-enhancing effects, welfare may be increased through the provision of public goods, such as libraries, parks, and art museums. This study investigates the growth and welfare implications of such public policies. In this endogenous growth model, we include the provision of public infrastructure, the provision of utility enhancing services, and allow for transfer payments as well as including distortionary taxes. We systematically examine the effects of alternative ways in which the government collects revenues and allocates its revenues among transfers, public investment, and expenditures on goods and services on growth and welfare.

Several papers have used endogenous growth models to analyze the effects of distortionary taxes on growth and welfare. A partial list of papers includes: King and Rebelo (1990), Lucas Jr. (1990), Jones et al. (1993), Pecorino (1993), and Cassou and Lansing (1997). In these models, the long-run growth rate is potentially reduced due to distortionary taxes that lower the incentives to work and save. The magnitude of the growth effects in these models ranges from almost zero (Lucas, 1990) to a fourfold increase (a special case analyzed in Jones et al., 1993). Stokey and Rebelo (1995) review this literature. By conducting experiments on simulated economies, Stokey and Rebelo (1995) show that large growth effects of fiscal policy occur when depreciation rates are implausibly large and/or when the uncompensated labor elasticity is implausibly high. In their view, one aspect of these artificial economies that does not seem to matter for growth is the elasticity of substitution among inputs, an issue we address below.

Most of the models cited above focus solely on the effects of taxation and ignore the potential benefits of public policies. Typically, tax revenues are either

rebated lump-sum to the households or are simply “thrown in the ocean”. Notable exceptions are Barro (1990), Glomm and Ravikumar (1994), and Turnovsky and Fisher (1995). In Barro and Glomm and Ravikumar, flat rate taxes are levied on income and tax revenues are used to fund public goods and services that enhance private returns. A tension then arises between the distortionary effects of taxation (lowering growth) and the increase in publicly provided investment (enhancing growth). In Turnovsky and Fisher (1995), the government collects lump-sum taxes and allocates these revenues to utility augmenting services and to infrastructure investment. They look at the welfare implications of different policies in a neoclassical growth model where the economy converges to a steady state.

Our model includes public capital, transfer payments, and allows for government expenditures on goods and services to enter into the utility function. Therefore, the model is closest in spirit to Barro (1990), Glomm and Ravikumar (1994), Turnovsky and Fisher (1995). However, our model differs from these models in four important ways. First, we allow the elasticity of substitution between private inputs and publicly provided inputs to differ from unity. Second, we include elastic labor supply.¹ Third, we model the accumulation process of physical capital, human capital, and public capital separately. Fourth, government revenues are used for lump-sum transfers and for services that the household may value as well as for public investment. The model here differs from those analyzed in Stokey and Rebelo (1995) since we include productive government capital. Therefore, we can generalize their results for elastic labor to incorporate the effects when public capital is included. More importantly, in Stokey and Rebelo (1995), the elasticity of substitution between private factors is not quantitatively important. We wish to generalize their result to examine the importance of the elasticity of substitution when one of the inputs is publicly provided.

Consequently, our model yields insights into the growth effects and welfare implications of public policy beyond what is obtainable in the framework used in Barro (1990), Glomm and Ravikumar (1994), Stokey and Rebelo (1995), and Turnovsky and Fisher (1995). First, the qualitative and quantitative growth and welfare effects of public policy depend on the elasticity of substitution between private and public capital. Second, unlike Barro and Glomm and Ravikumar, since we model the accumulation decision of the capital stocks separately, we can allow for different tax rates on physical and human capital. Third, the model is capable of quantifying the welfare effects from altering the allocation of revenues between growth enhancing public investment and utility augmenting

¹ Turnovsky and Fisher do include elastic labor supply. However, since their analysis is restricted to lump-sum taxes, they do not investigate the effects of distortionary taxes on growth and welfare.

public services. For plausible parameters, we find that welfare is maximized when the majority of tax revenues are allocated to public investment; that is, the positive welfare effects from higher growth rates dominate the static welfare effects from increases in publicly provided services. Though the growth effects from changing the size of the government are, in general, modest, we find that the welfare implications can be significant.

In the end, three arguments emerge with respect to public policies. First, the growth and welfare consequences of increasing public infrastructure investment can be substantial. Significant welfare gains can be obtained even if the increased infrastructure investment comes at the expense of a reduction in the provision of utility augmenting public services. Second, with a balanced government budget, taxing profits at a higher rate than labor income raises long-run growth. In the context of this model this occurs because profits include public capital's rents and so profits should be taxed at a higher rate than wage income. Also, we show that taxes on profits should be higher than labor taxes when public capital and private capital are less substitutable. This occurs because as the elasticity of substitution between public capital and private capital falls the growth maximizing ratio of public capital to private capital increases. A higher tax rate on profits and a lower tax rate on labor income achieves this higher ratio of public capital to private capital. Finally, as in Stokey and Rebelo (1995), the growth effects of public policies are magnified the more elastically labor is supplied.

The rest of the paper proceeds as follows. Section 2 describes the general model. Section 3 examines the growth effects of public policies. We first consider a special case where closed-form solutions can be obtained. We then relax the assumptions of inelastically supplied labor and unitary elasticity of substitution between private and public inputs to qualitatively and quantitatively assess the long-run growth effects of public policies. Section 4 examines the welfare implications including transitional dynamics. In this section, we assess the welfare effects from changes in the composition of government spending and from changes in the relative size of the government. Section 5 concludes.

2. The model

The economy is populated by a large number of infinitely lived households. Private agents (households and firms) take fiscal policies as given when making private optimal decisions. The government levies taxes on labor income and gross profits of the firm. Revenues from taxation are used for public investment, providing public services, and for transfer payments. In the following subsections, we describe the actions of the households, firms, and the government in greater detail.

2.1. Households

We assume all households are identical and that there is no population growth. We normalize the population size to unity and focus on the actions of a representative household. Preferences of the representative household are given by the utility function:

$$U = \sum_{t=0}^{\infty} \frac{\beta^t (c_t^\mu (S_t)^{1-\mu})^{1-\sigma} (1-n_t)^\varepsilon}{(1-\sigma)}, \quad \begin{aligned} &0 < \mu \leq 1, \\ &\sigma > 1 \Rightarrow \varepsilon < 0, \\ &0 < \sigma \leq 1 \Rightarrow 0 < \varepsilon < 1, \\ &0 < \beta < 1, \end{aligned} \quad (1)$$

where c_t is consumption at time t , n_t is work effort at time t , with a unit time endowment $(1 - n_t)$ is leisure at time t , S_t represents services provided by the government which enhance utility, and β is the household's rate of time preference.²

In each period, the household supplies effective labor ($h_t n_t$), where h_t is the level of human capital at time t . The household receives the wage (w_t) for each unit of effective labor supplied. Labor income is taxed at the flat rate τ_L . In addition to labor income, the household receives dividend payments from the firm (D_t), and transfer payments from the government (T_t). In some cases, we will permit households to 'expense' depreciated human capital ($\delta_h h_t$), where δ_h is the depreciation rate of human capital.³

Households use their income to purchase consumption goods and accumulate human capital. The lifetime budget constraint of the household is given by

$$\begin{aligned} &\sum_{t=0}^{\infty} p_t \{ (1 - \tau_L)(w_t h_t n_t) + \tau_L \eta_H \delta_H h_t + D_t + T_t \} \\ &= \sum_{t=0}^{\infty} p_t (c_t + h_{t+1} - (1 - \delta_h) h_t), \end{aligned} \quad (2)$$

where p_t is the price of the time t good, and $\eta_H \in \{0, 1\}$ is the extent to which depreciated human capital can be expensed. If $\eta_H = 1$ ($\eta_H = 0$) depreciated

² Such a utility function without government services has been used by King et al. (1988). Along the growth path, this specification yields a constant fraction of time allocated to work effort. We use lower case letters to denote the household and firm's choices and uppercase letters to denote aggregate levels of publicly provided goods.

³ As in Frenkel and Razin (1996) expensing human capital depreciation does not exist in this exact form, but it can be thought of as mimicking the effects of tax progressivity and subsidized health care.

human capital can (cannot) be expensed. Given an initial endowment of human capital (h_0), a sequence of prices $\{p_t, w_t, \}_{t=0}^{\infty}$, and public policies $\{T_t, S_t, \tau_L\}_{t=0}^{\infty}$ the household maximizes (1) subject to (2).

2.2. Firms

Each household operates a goods producing firm.⁴ Private capital, public capital and effective labor time are inputs into the production process. More specifically, technology is given by

$$y_t = A[\varphi G_t^\rho + (1 - \varphi)k_t^\rho]^{\alpha/\rho} (h_t^d n_t^d)^{1-\alpha},$$

$$0 < \alpha < 1, \quad 0 < \phi < 1, \quad -\infty < \rho \leq 1, \quad (3)$$

where $h_t^d n_t^d$ is effective labor demand, G_t is public capital, $(1 - \alpha)$ is labor's share of output, and ρ measures the elasticity of substitution between public and private physical capital. In this formulation, there are constant returns to scale in augmentable factors to allow for constant growth. We have allowed for varying degrees of substitutability between public and physical capital. The evidence on substitutability between public capital and private capital is mixed. Nadiri and Mamuneas (1994), for example, find high elasticity of substitutability between public and physical capital and Berndt and Hansson (1991) find evidence of low elasticity of substitution. We begin with this general CES specification in order to accommodate different cases and quantitatively and qualitatively assess the importance of the elasticity of substitution for long-run growth and welfare.

The firm pays taxes (τ_k) on gross profits, and uses the after-tax profits to finance capital purchases. We define gross profits at time t as

$$\Pi_t = [A\varphi G_t^\rho + (1 - \varphi)k_t^\rho]^{\alpha/\rho} (h_t^d n_t^d)^{1-\alpha} - w_t h_t^d n_t^d - \eta_K \delta_k k_t,$$

where δ_k is the depreciation rate on physical capital and $\eta_K \in \{0, 1\}$ measures the extent to which depreciation of physical capital can be expensed. Dividends are paid out net of capital purchases and are given by

$$D_t = (1 - \tau_k) \{ [A\varphi G_t^\rho + (1 - \varphi)k_t^\rho]^{\alpha/\rho} (h_t^d n_t^d)^{1-\alpha} - w_t h_t^d n_t^d \} \\ - k_{t+1} - (1 - \delta_k (1 - \tau_k \eta_K)) k_t.$$

Since firms are owned by the household, a natural assumption is to assume that firms behave in the best interest of their owners. Thus, the firm values a dividend dollar in period t by the marginal utility of a household at time t .

⁴ A similar market structure is employed by Lansing (1998) to examine fiscal policies at business cycle frequencies.

Therefore, taking as given the sequence of prices, wages, and public policies, the firm chooses a sequence of (h_{t+j}^d, n_{t+j}^d) and k_{t+j+1} to maximize

$$\sum_{j=0}^{\infty} \beta^j u'(c_{t+j}) D_{t+j}$$

where $u'(c_{t+j})$ represents the marginal utility of time $t + j$ private consumption.⁵

2.3. Public sector

The role of the public sector in our model is to collect taxes and use the revenues for transfer payments, infrastructure investment, and utility enhancing services. As in Barro (1990), Lucas (1990), Pecorino (1993), and Glomm and Ravikumar (1994), we assume the government maintains a balanced budget in all periods; that is,

$$\tau_L w_t h_t n_t + \tau_k \Pi_t - \tau_L \eta_H \delta_h h_t = T_t + S_t + I_{G,t},$$

where $I_{G,t}$ is government investment at time t .⁶ We assume that all revenues are raised through distortionary taxes and that transfer payments are non-negative.

Letting m_1 be the fraction of government revenues that goes to public investment at time t , we can express the accumulation equation for public capital as

$$G_{t+1} = m_1 \{ [\alpha \tau_k + (1 - \alpha) \tau_L] y_t - (\tau_L \eta_H \delta_h h_t + \tau_k \eta_K \delta_k k_t) \} + (1 - \delta_g) G_t,$$

where δ_g is the depreciation rate of public capital. Letting m_2 be the fraction of government revenues that goes to government services, transfer payments at time t are given by

$$T_t = [1 - m_1 - m_2] (\tau_L w_t h_t n_t + \tau_k \Pi_t - \tau_L \eta_H \delta_h h_t).$$

2.4. Equilibrium

Given a fiscal policy, a competitive equilibrium is a set of allocations $\{c_t, n_t, h_{t+1}, k_{t+1}\}_{t=0}^{\infty}$ together with prices $\{p_t, w_t\}_{t=0}^{\infty}$ such that:

- (i) $\{c_t, n_t, h_{t+1}\}$ solves the household's problem,
- (ii) $\{n_t^d, h_t^d, k_{t+1}\}$ solves the firm's problem,

⁵ Christiano (1991) uses a similar objective function in studying the effects of monetary policies.

⁶ Saint-Paul (1992) studies the growth implications of public deficit.

- (iii) $h_t n_t = h_t^d n_t^d$, that is, the labor market clears,
- (iv) the household's budget constraint is satisfied:
 $(1 - \tau_L)w_t h_t n_t + D_t + T_t = c_t + h_{t+1} - (1 - \delta_h(1 - \tau_L \eta_H))h_t$
- (v) the government's budget constraint is satisfied:
 $\tau_L w_t h_t n_t + \tau_k \Pi_t - \tau_L \eta_H \delta_h h_t = T_t + S_t + I_{G,t}$.

The competitive equilibrium yields the following first-order conditions: from the firm's problem, the real wage is equal to the marginal product of effective labor

$$w_t = (1 - \alpha)y_t/h_t n_t$$

and the Euler equation for physical capital accumulation is given by

$$\begin{aligned} & \frac{\mu(c_t^\mu (S_t^{1-\mu})^{1-\sigma})^{1-\sigma}}{c_t} (1 - n_t)^\varepsilon \\ &= \beta \frac{\mu(c_{t+1}^\mu S_{t+1}^{1-\mu})^{1-\sigma} (1 - n_{t+1})^\varepsilon}{c_{t+1}} \left\{ \frac{(1 - \tau_k)\alpha(1 - \varphi)k_{t+1}^\rho}{\varphi G_{t+1}^\rho + (1 - \varphi)k_{t+1}^\rho} \frac{y_{t+1}}{k_{t+1}} \right. \\ & \quad \left. + (1 - \delta_k(1 - \tau_k \eta_K)) \right\}. \end{aligned} \quad (4)$$

The household chooses work effort so that marginal rate of substitution between work effort and consumption is equal to the marginal product of raw labor

$$\frac{\varepsilon}{\mu(1 - \sigma)} \frac{c_t}{1 - n_t} = (1 - \tau_L)(1 - \alpha)y_t/n_t \quad (5)$$

and the Euler equation for human capital accumulation is given by

$$\begin{aligned} & \frac{\mu(c_t^\mu S_t^{1-\mu})^{1-\sigma}}{c_t} (1 - n_t)^\varepsilon \\ &= \beta \frac{\mu(c_{t+1}^\mu S_{t+1}^{1-\mu})^{1-\sigma}}{c_{t+1}} (1 - n_{t+1})^\varepsilon \{ (1 - \alpha)(1 - \tau_L)y_{t+1}/h_{t+1} \\ & \quad + (1 - \delta_h(1 - \tau_L \eta_H)) \}. \end{aligned} \quad (6)$$

Recall the transition equation for public capital accumulation is given by

$$G_{t+1} = m_1 \{ [\alpha \tau_k + (1 - \alpha)\tau_L]y_t - (\tau_L \eta_H \delta_h h_t + \tau_k \eta_K \delta_k k_t) \} + (1 - \delta_g)G_t. \quad (7)$$

and the expression for public services are given by

$$S_t = m_2 \{ (\alpha \tau_k + (1 - \alpha)\tau_L)y_t - (\tau_L \eta_H \delta_h h_t + \tau_k \eta_K \delta_k k_t) \}. \quad (8)$$

Using Eqs. (4)–(8), the remainder of the paper assesses the effects of public policies on growth and welfare.

3. Long-run growth effects of public policy

Along the balanced growth path, output, consumption, and all capital stocks grow at the same rate, work effort is constant, and the ratios c_t/k_t , h_t/k_t , G_t/k_t are constant for all t . This implies that along the balanced growth path equations (4)–(8) yield a system of five equations for the five unknowns ($\gamma, n, c/k, G/k, h/k$) where γ is the gross growth rate of output, consumption, and the three capital stocks.

The five equations are

$$\frac{\varepsilon}{\mu(1-\sigma)} \frac{n}{1-n} = \frac{(1-\tau_L)(1-\alpha)A[\varphi(G/k)^\rho + (1-\varphi)]^{z/\rho}(h/k)^{1-\alpha}n^{1-\alpha}}{c/k}, \quad (9)$$

$$\begin{aligned} & \frac{\gamma^\sigma - \beta(1-\delta_k(1-\tau_k\eta_K))}{\alpha\beta(1-\tau_k)} \\ &= \frac{(1-\varphi)}{\varphi(G/k)^\rho + (1-\varphi)} A\{\varphi(G/k)^\rho + (1-\varphi)\}^{z/\rho}(h/k)^{1-\alpha}n^{1-\alpha}, \end{aligned} \quad (10)$$

$$\frac{\gamma^\sigma - \beta(1-\delta_h(1-\tau_L\eta_H))}{(1-\alpha)\beta(1-\tau_L)} = A\{\varphi(G/k)^\rho + (1-\varphi)\}^{z/\rho}\left(\frac{h}{k}\right)^{-\alpha}n^{1-\alpha}, \quad (11)$$

$$\begin{aligned} & \frac{\gamma - (1-\delta_g) - \tau_L\eta_H\delta_h h/k - \tau_k\eta_K\delta_k}{m_1(\alpha\tau_k + (1-\alpha)\tau_L)} \\ &= \left[A\{\varphi(G/k)^\rho + (1-\varphi)\}^{z/\rho}\left(\frac{h}{k}\right)^{1-\alpha}n^{1-\alpha} \right] \left(\frac{G}{k}\right)^{-1}, \end{aligned} \quad (12)$$

$$\begin{aligned} c/k &= \{1 - (m_1 + m_2)(\alpha\tau_k + (1-\alpha)\tau_L)\} \\ &\quad \times A\{\varphi(G/k)^\rho + (1-\varphi)\}^{z/\rho}\left(\frac{h}{k}\right)^{1-\alpha}n^{1-\alpha} \\ &\quad - (\gamma - (1-\delta_k(1-\eta_K\tau_k))) - (\gamma - (1-\delta_h(1-\eta_H\tau_L)))(h/k) \\ &\quad + (m_1 + m_2 - 1)(\eta_K\delta_k\tau_k + \eta_H\delta_h\tau_L)(h/k). \end{aligned} \quad (13)$$

We begin by looking at the special case where the production function is Cobb–Douglas and leisure is not valued. In this case, some closed-form solutions can be obtained. This case will serve as a benchmark and comparisons can

be made when we allow leisure to be elastically supplied and when we allow for the substitutability of private and public capital to differ from unity.

3.1. Cobb–Douglas technology, inelastic leisure, and no expensing of depreciated capital

To determine the growth effects of public policy we need only consider Eqs. (10)–(12). These three equations can be substituted into (3) to yield

$$1 = A \left[\frac{m_1(\alpha\tau_k + (1-\alpha)\tau_L)}{\gamma - (1-\delta_G)} \right]^{\alpha\varphi} \left[\frac{\alpha(1-\varphi)\beta(1-\tau_k)}{(\gamma^\sigma - \beta(1-\delta_k))} \right]^{\alpha(1-\varphi)} \times \left[\frac{(1-\alpha)\beta(1-\tau_L)}{\gamma^\sigma - \beta(1-\delta_h)} \right]^{(1-\alpha)}. \quad (14)$$

Given (14) it is straightforward to assess the effects of public policies on long-run growth. The following propositions succinctly show the interactions between public policies and growth. Taking the logs of both sides of (14) and totally differentiating, we obtain:

*Proposition 1. With m_2 , τ_L , and τ_k held constant, the growth rate is monotonically increasing in infrastructure investment.*⁷

Notice that in Proposition 1 we are holding fixed both tax rates. This essentially fixes total tax revenues. We are also holding constant the fraction of the government revenues allocated to utility enhancing services. The policy experiment then is to increase the fraction of the budget going to infrastructure investment at the expense of lump-sum transfers. The proposition shows that increasing (decreasing) infrastructure investment (lump-sum transfers) increases (decreases) the growth rate. The intuition behind this result is straightforward: increasing the fraction of output that goes to public investment raises the stock of public capital which increases the marginal product of physical and human capital which, in turn, increases growth. Alternatively, reducing transfer payments has no direct effect on physical capital and human capital accumulation so increasing infrastructure investment and reducing transfer payments increases growth.⁸

⁷ See the appendix for derivations of Propositions 1–3.

⁸ It is easy to show that the growth rate is a concave function of public investment. This occurs because of the second-order effects of reducing transfer payments. All else held constant, by reducing transfer payments, the representative household allocates fewer resources to physical capital and human capital accumulation. This mitigates the positive growth effects from increased infrastructure investment.

Proposition 2. For fixed m_1, m_2 and for $\tau_k = \tau_L = \tau$, the tax rate that maximizes growth is given by

$$\tau = \alpha\varphi.$$

The intuition behind this result is the same as in Barro (1990): the marginal social benefit of public capital accumulation is equated to the marginal social cost of public capital accumulation.

Even in this simplified setting, one of the novel features of this model that differentiates the model from Barro (1990) and Glomm and Ravikumar (1994) is that we can allow the tax rates to vary on human and physical capital and investigate the growth effects of a model with public capital when we allow for different tax rates on the inputs. To this end, we define government's share of output $s = [\alpha\tau_k + (1 - \alpha)\tau_L]$ and derive the growth maximizing tax rates on human and physical capital when s is held fixed. This leads to the following proposition.

Proposition 3. For fixed m_1, m_2 , and s the tax rates that maximize the growth rate are⁹

$$\tau_k = \lfloor (1 - \varphi)s + \varphi(1 - \alpha) \rfloor / (1 - \alpha\varphi), \quad (15)$$

$$\tau_L = s - \alpha\varphi. \quad (16)$$

Proof. This follows from substituting $\tau_L = (s - \alpha\tau_k)/(1 - \alpha)$ into (14) and totally differentiating (14) and solving for τ_k . $\square$

Proposition 3 leads to the following corollary.

Corollary 1. The growth maximizing tax rates on physical capital and human capital are given by

$$\tau_k = \varphi, \quad \tau_L = 0.$$

Proof. This follows from substituting for $s = \alpha\tau_k + (1 - \alpha)\tau_L$ into (15) and (16) and solving for τ_L and τ_k . $\square$

Pecorino (1993) shows that when labor is inelastically supplied the growth maximizing tax rates on human and physical capital are equal. In our framework, this is not the case. This is due to the fact that dividend payments include

⁹ Here we assume that s is greater than or equal to $\alpha\varphi$.

pure profits or, perhaps more appropriately stated, public capital's 'rents'.¹⁰ These rents arise only because of the public provision of a productive input. Therefore, the growth maximizing tax rate on physical capital will be higher as the government taxes these rents.¹¹

Given the results above, if the goal of the fiscal authority is to maximize long-run growth then the following public policies will achieve this goal: $m_1 = (1 - m_2)$, $s = \alpha\varphi$, $\tau_k = \varphi$, $\tau_L = 0$.

3.2. Cobb–Douglas technology, elastic leisure, and expensing of depreciated capital

In this section, we generalize the model to allow for the case where leisure is valued by the household. In addition, we assume the depreciation of physical and human capital stocks can be expensed. The inclusion of valued leisure implies that we solve Eqs. (9)–(13) with $\rho = 0$. In models where the government's sole role is to tax (and not to provide productive inputs), Stokey and Rebelo (1995) show that incorporating elastically supplied labor can have important growth effects. We wish to see how incorporating elastically supplied labor affects growth for models when public capital is an input into the production process. Once the leisure–labor choice is incorporated, closed-form solutions cannot be obtained so the model must be calibrated and simulated. Therefore, to assess the effects of public policy on long-run growth we simulate the model conducting policy experiments which roughly correspond to Propositions 1–3.

3.2.1. Calibration

We choose our parameters to match the average, annual post-war experience for the US. We set the gross growth rate of output $\gamma = 1.018$; this implies the balanced growth rate of output is equal to 1.8% per year. For the inter-temporal elasticity of substitution we set $\sigma = 2$ (as in Lucas, 1990). The household's subjective rate of time preference (β) is chosen to be 0.98. This implies an after-tax real rate of return of 5.7%. We experiment with different values of ε ranging from $\varepsilon = 0$ to -1.5 .

¹⁰ In some respects the results are similar to the results in Jones et al. (1997). In their model, the limiting tax rate on capital was non-zero. This arises in their framework because government capital augments infrastructure investment. In this case, it is optimal to tax the rents that arise from enhancing investment projects.

¹¹ In the limiting case where $\varphi = 0$, the government would collect no taxes and allocate nothing to infrastructure investment, publicly provided services, and transfer payments. In most cases, this does not create any problems. The only instance a problem arises is when publicly provided services are valued. If no revenues were allocated to services, the utility function of the household would not be well defined. To circumvent this issue we have restricted our analysis to the case where φ is strictly positive.

For the technology parameters, labor's share of output in the US has remained at about 65% in the post-war period. This implies we set $(1 - \alpha) = 0.65$. We choose $\varphi = 0.30$. This implies the elasticity of output with respect to public capital is 0.105 a number within the bounds of Garcia-Mila and McGuire (1992), Lynde and Richmond (1993), Holtz-Eakin (1994), and Ai and Cassou (1995). We jointly determine the depreciation rates of physical capital and public capital so that the capital-output ratio is 2.45 (Lucas, 1990) and the public capital–physical capital ratio is 0.23 (Lansing, 1998). Given our other parameters, this implies a depreciation rate of 0.04 on physical capital and a depreciation rate of 0.063 on public capital. For simplicity, we choose the depreciation rate on human capital to be the same as the depreciation rate on private physical capital. This yields a human capital–private, physical capital ratio of 2.52. Finally, we set the technology parameter A , so output growth is 1.8% per year along the balanced growth path.

Regarding the public policy parameters, we set the tax rates on labor income and profits to be 35%. These tax rates are slightly lower than the values used in Lucas (1990) and slightly higher than the values used in Lansing (1995). To calculate the percent of revenues allocated to infrastructure investment, we know total public investment in the US in 1992 (\$171.1 billion) as a share of national output (\$5.1 trillion) was 3.4%; given government's share of output we choose $m_1 = 0.14$. We choose m_2 so that total expenditures the government ($m_1 + m_2$) is equal to 22% of GDP (Lucas, 1990). This implies $m_2 = 0.40$.

3.2.2. Policy experiments

The first policy experiment we conducted corresponds to Proposition 1. For given tax rates on labor income and profits ($\tau_k = \tau_L = 0.35$) and a fixed allocation of revenues to publicly provided goods and services ($m_2 = 0.40$), we examine the growth effects from increasing the percent of revenues allocated to public investment. Figs. 1a–c illustrate these growth effects. In all cases the growth rate monotonically increases in m_1 . The intuition for this is the same as in the previous section: increasing the percent of revenues that goes to infrastructure investment increases the stock of public capital which increases the marginal products of human and physical capital which raises the growth rate. When $\varepsilon = 0$, the growth rate ranges from -0.02 to 2.54% . As ε increases in absolute value, the graph of output growth as a function of infrastructure investment becomes steeper. The intuition for this is straightforward. As the stock of public capital increases due to more public investment, the marginal product of raw labor increases. With a more elastic labor supply schedule (ε larger in absolute value), the response in work effort is greater. The increase in work effort increases the marginal product of physical capital and human capital which, in turn, increases the growth rate. When $\varepsilon = -0.50$ (as in Lucas, 1990) output growth ranges from -0.04 to 2.84% . When $\varepsilon = -1.50$ output growth ranges from -0.06 to 3.12% .

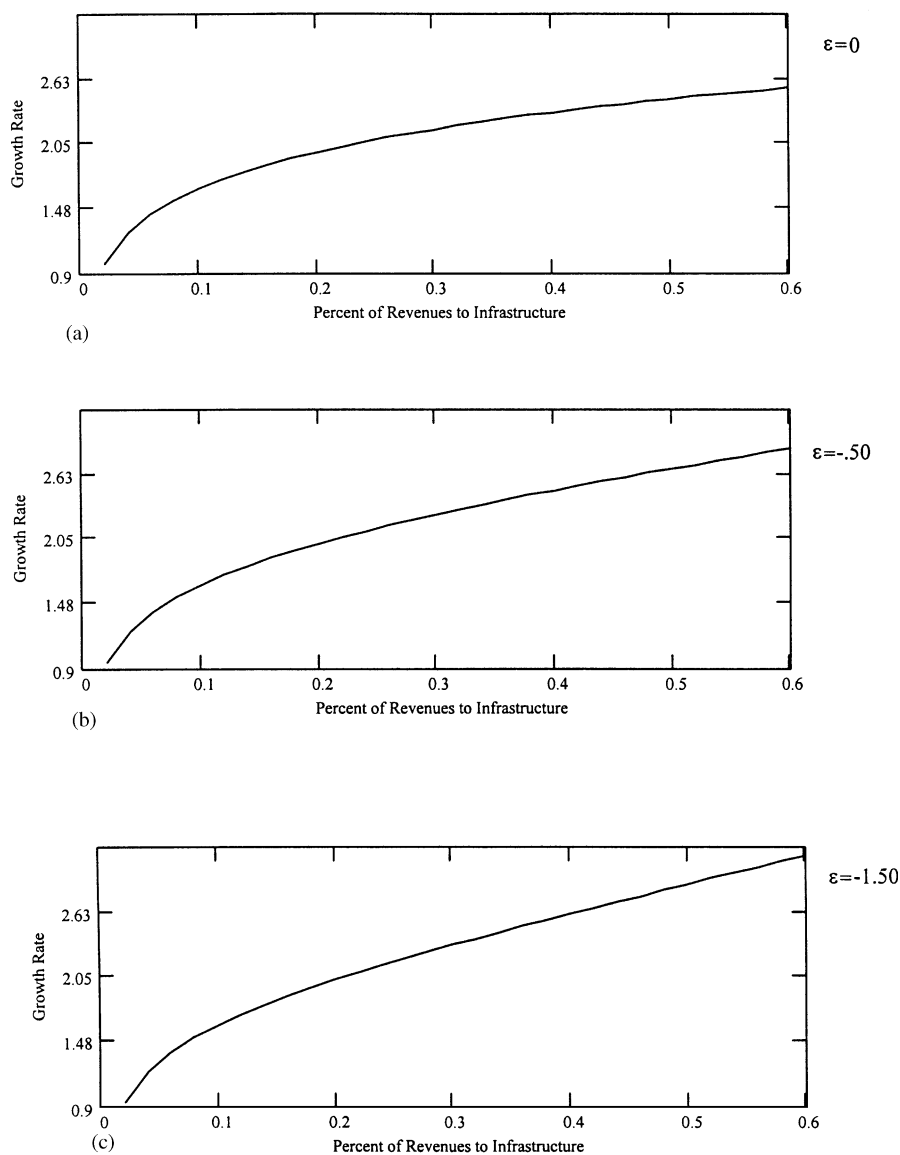


Fig. 1. Growth rate as a function of the percent of revenues allocated to infrastructure investment.

Our second experiment investigates the effects on growth when the government's share of output varies. By simply adding the leisure–labor choice, we show that the implications for the growth maximizing size of the government is qualitatively and quantitatively different than Barro (1990) and Glomm and

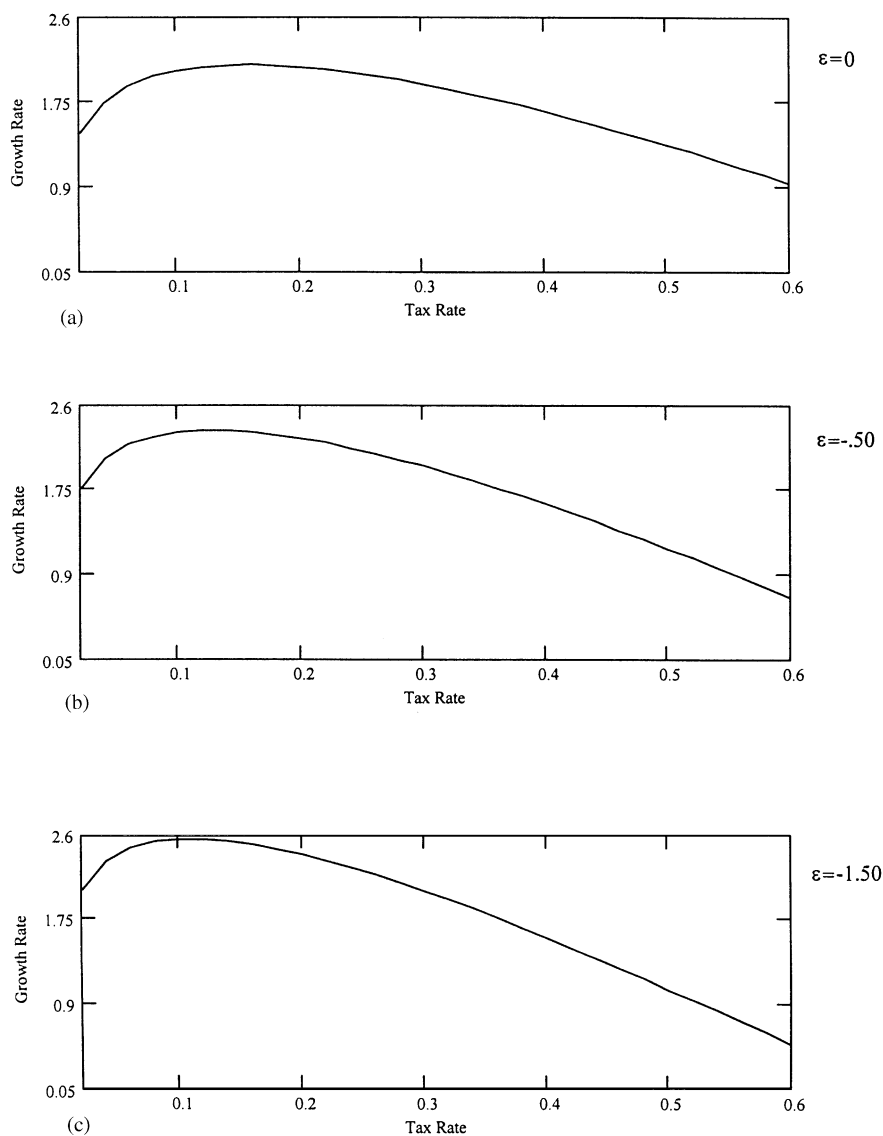


Fig. 2. Growth rate as a function of the government's share of output.

Ravikumar (1994). As in Barro and Glomm and Ravikumar, we set $\tau = \tau_k = \tau_L$ and vary τ .

The growth effects that occur from increasing government's share of output are depicted in Fig. 2. As in Barro and Glomm and Ravikumar, the growth rate

is a non-monotonic function of government's share of output. The inverted U-shape occurs because low tax rates imply there is under-investment in infrastructure. This results in low growth. As the size of the government increases, infrastructure investment increases. The accumulation of public capital raises the marginal product of human and physical capital which raises the growth rate. At some point, the distortionary effects of taxation slows the accumulation of private inputs and the growth rate falls. The growth rate is no longer maximized when the tax rate is equal to the elasticity of output with respect to public capital. Two reasons account for this: first, not all tax revenues go to infrastructure investment. Since tax revenues are used to fund transfer payments, goods and services, as well as infrastructure investment, more revenues are needed to provide an 'adequate' level of public capital. Accordingly, the growth maximizing tax rate will be higher. Second, output growth is sensitive to ε . When $\varepsilon = 0$, output growth is maximized when government's share is 16%. When labor supply is still relatively inelastically supplied ($\varepsilon = -0.50$), the growth rate is maximized when government's share of output is 12%. When $\varepsilon = -1.50$ the growth rate peaks when government's share is 10%. As before, the effects of public policy are magnified when the elasticity of labor supply increases. More specifically when government's share of output is low, an increase in the tax rate raises the stock of public capital. This increases the marginal product of raw labor. The more elastic the labor supply schedule the more employment increases. The more employment increases the greater the increase in the marginal products of human and physical capital. The higher marginal products raise the growth rate. Conversely, when government's share of output is too high, an increase in the tax rate decreases the incentives to work (because workers choose to consume more leisure). Therefore, the more elastic the labor supply schedule the greater the fall in employment. This drop in employment lowers the marginal products of human and physical capital which lowers the growth rate.

Finally, we analyze numerically the analog of Proposition 3. In a model without infrastructure investment and inelastic leisure, Pecorino (1993) shows that the growth is maximized when the tax rates on labor income and capital income are equal. If labor is elastically supplied, Pecorino shows that the growth rate is maximized when capital income is taxed at a higher rate than labor income.

Fig. 3 depicts the results for different values of ε . In all cases, the growth maximizing tax rate on labor income is less than the tax rate on profits. In the case where labor is supplied inelastically, growth is maximized when labor income (profits) are taxed at the rate 22.9% (51.4)%. The intuition is similar to before: Since profits include the government's rents, profits should be taxed at a higher rate. As ε increases in absolute value, the growth maximizing tax rate on labor income falls. The intuition for this is the same as in Pecorino (1993); that is, as labor supply becomes more elastically supplied, taxes on labor income

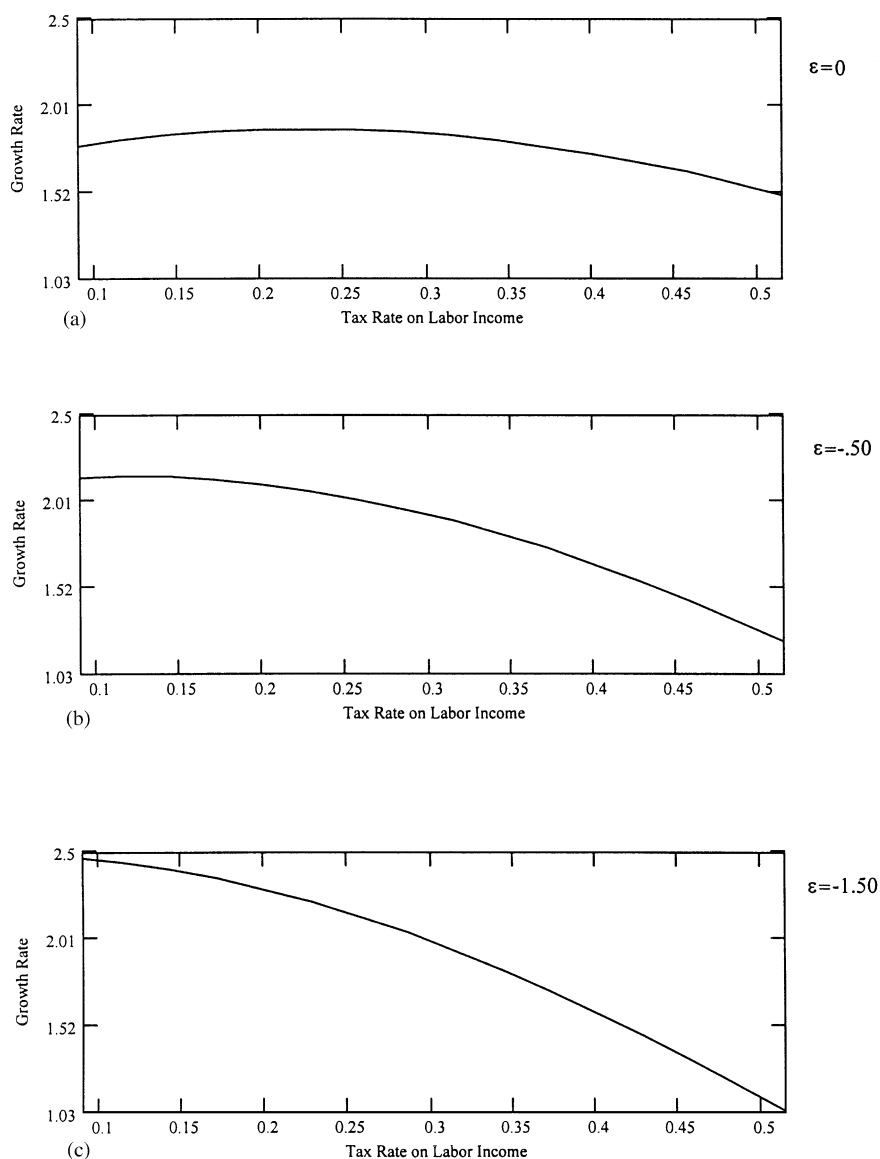


Fig. 3. Growth rate as a function of labor taxes holding government's share constant.

become more distortionary. Accordingly, the growth maximizing tax on labor income is lower. It is interesting to note that when $\varepsilon = -1.50$ (a value frequently employed in the real business cycle literature) the growth maximizing tax rate on labor income is near zero (2.9%). We have done sensitivity analysis

with respect to φ . Similar to the case where closed-form solutions were obtained we find that: as φ increases (decreases), the growth effects from increasing infrastructure investment are larger (smaller). Also as φ increases (decreases), the growth maximizing tax rate on profits and labor income is higher (lower). Finally, as φ increases (decreases), the growth maximizing tax on profits relative to the tax on labor income increases (decreases).¹²

3.3. General CES technology

In this section we investigate the effects of public policies when the elasticity of substitution between private inputs and public inputs differs from unity. In the context of our model, this involves varying the parameter ρ .¹³ There are some examples where we would expect a high degree of complementarity (e.g. highways and trucking) and some examples where we would expect to see a high degree of substitutability (trucks owned by the US postal services and UPS trucks). Empirically, however, there is little known concerning the degree of substitutability between private and public capital. Nadiri and Mamuneas (1994) find a high degree of elasticity of substitution, while Berndt and Hansson (1991) find a low degree of substitution. Thus, the degree of substitutability is an open question. We experiment with a variety of values for ρ and investigate whether the degree of substitutability matters for growth.

Is there reason to believe that the elasticity of substitution matters for growth? Stokey and Rebelo (1995) show that the elasticity of substitution among *private* inputs is not critical for determining growth. Even though changes in tax rates affect input ratios, Stokey and Rebelo show that for empirically plausible parameters the after-tax marginal products of human and physical capital are insensitive to the elasticity of substitution. All else held constant, if the after-tax marginal products are insensitive to tax policies, the growth rate will be as well. We attempt to extend their analysis to include the case where one of the inputs is publicly provided and the degree of substitutability differs from unity. In the context of our model, one obvious conjecture is that if the degree of substitutability between public capital and private capital is high, a public policy that increases (decreases) the stock of public capital will result in private agents decreasing (increasing) the stock of physical capital. Since the decrease (increase) in physical capital offsets the increase (decrease) in public capital, the after-tax

¹² The analog to Corollary 1, in this case, is that when $\varepsilon = 0$ and without expensed depreciation the growth maximizing size of the government is 10.5% of output and all taxes should be levied on profits. As ε increases in absolute value, the positive growth effects from taxing dividends are reinforced.

¹³ There are alternative formulations with varying degrees of substitutability between public and human capital. We focus on the technology described in (3).

marginal product of human capital is unaffected. If the marginal product of human capital is unaffected there will be no change in the growth rate. Thus, consistent with Stokey and Rebelo (1995), high degrees of substitutability between inputs should result in negligible growth effects. However, if the degree of substitutability between public capital and private capital is low, then a public policy that increases (decreases) the stock of public capital may raise (lower) the marginal product of physical capital and lead to more (less) physical capital accumulation. If both public capital and physical capital stocks increase (decrease), the marginal product of human capital increases (decreases) and the growth rate increases (decreases). Therefore, when the degree of substitutability is low, public policies may have significant growth effects. We examine these conjectures more carefully below. We again conduct experiments that correspond to Propositions 1–3 to determine the extent to which the elasticity of substitution between public and private inputs matter for growth. All the parameters are the same as in Section 3.2, except we allow for different values of ρ . For our benchmark case, we set $\varepsilon = -0.50$.

The first experiment is to increase the percent of revenues allocated to infrastructure investment. Fig. 4 depicts the results when $\rho \in \{-0.85, -0.50, 0.25, 0.75\}$. When public capital and private capital are near substitutes as in Fig. 4a, the growth effects from increasing the percent of government revenues to infrastructure investment are rather small. The growth rate ranges from 1.72% (when $m_1 = 0.02$) to 2.14% (when $m_1 = 0.60$). Two effects cause the growth rate to increase. First, since private capital and public capital are not *perfect* substitutes, the increase in infrastructure investment raises the marginal products of human capital and physical capital which causes output growth to rise. Second, increasing the percent of revenues to infrastructure investment increases the marginal product of raw labor. This causes work effort to increase. The increase in raw labor raises the marginal products of human capital and physical capital which also causes output growth to rise.¹⁴ When public capital and physical capital are less substitutable the growth effects from increasing the percent of revenues to infrastructure investment are more pronounced. When $\rho = -0.85$ as in Fig. 4d, the growth rate ranges from -0.76% (when $m_1 = 0.02$) to 4.64% (when $m_1 = 0.60$).¹⁵ This occurs because increasing the stock of public capital raises the marginal products of physical capital and human capital which results in higher growth. Thus, the conjecture concerning the elasticity of substitution between private capital and public capital holds for infrastructure

¹⁴ If labor was inelastically supplied ($\varepsilon = 0$), then a policy that increases the percent of revenues from infrastructure investment from 14% (the benchmark case) to 60% would increase the growth rate from 1.8% to only 1.96% (an 8.8% increase).

¹⁵ The elasticity of output growth switching the current rate of public investment to the growth maximizing rate of public investment ($m_1 = 0.60$) is 0.48.

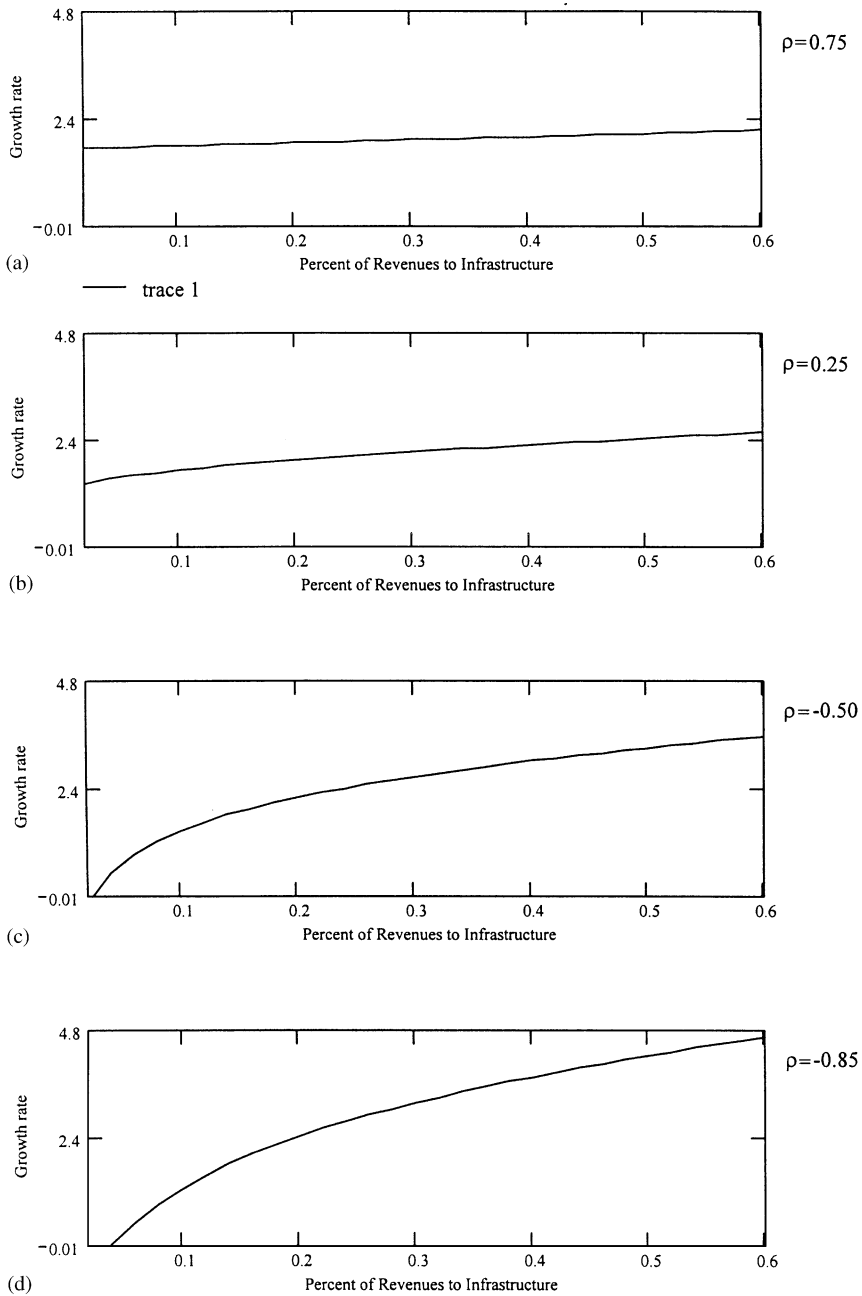


Fig. 4. Growth rate as a function of the percent of revenues to infrastructure investment.

investment; that is, increasing the percent of revenues to infrastructure investment has larger growth effects when public capital and physical capital are less substitutable.

The second experiment is to increase the tax rates on labor income and profits to investigate the affect of increasing government's share of output on growth. The results are depicted in Fig. 5. When public capital and private capital are more substitutable ($\rho = 0.75$ and 0.25), the growth maximizing size of the government is small.¹⁶ The reason for this is if public capital and private capital are close substitutes, it would not matter what type of capital (public or private physical) is in place. However, in order to accumulate public capital, the government must levy taxes which distorts private agents' decisions to work and save. Consequently, growth is higher if most of the accumulation of capital stocks is left to the private sector.

When public capital and private capital are less substitutable as in Fig. 5c and d, the growth maximizing size of the public sector is larger. When $\rho = -0.50$ ($\rho = -0.85$), the tax rate that maximizes the growth rate is 26 (30)% and the growth rate is 1.92 (1.83)%.¹⁷ It is easy to show that the growth maximizing ratio of public capital to private capital decreases in ρ . Thus, for fixed m_1 , a lower elasticity of substitution between public capital and physical capital implies that growth maximizing size of the public sector is larger. For low values of ρ , this implies growth will be maximized with a larger government's share of output. It is interesting to note that when government's share of output is low the growth effects from increasing the size of the government are quite large. This is evident by the initial steepness of the graphs in Fig. 5c and d. At some point, the distortionary effects of taxation outweigh the positive effects of increased public capital and growth falls. However, as the government size becomes larger and the growth rate falls, the slope of graph is relatively flatter indicating that the effects of increased public capital are still important.

Fig. 6 illustrates the relationship between the growth rate of output and the tax rate on labor income when the profit tax adjusts to keep government's share of output constant. The growth rate peaks at lower tax rates on labor income for lower values of ρ . Fig. 6a–c have an inverted U-shape. When the tax rate on labor income is low, the profit tax is high and too little physical capital is accumulated and this results in low growth. As the tax rate on labor (profits) income increase (decreases) more physical capital is accumulated and the growth rate increases. At some point, the tax rate on labor income becomes too high and too little human capital is accumulated, and the growth rate falls.

¹⁶ When $\rho = 0.75$, the growth rate reaches a maximum of 3.84% with a tax rate of 0.00000009%. When $\rho = 0.25$, the growth rate reaches a peak of 2.99% with a tax rate of 4%.

¹⁷ The elasticity of output growth from switching the current tax rates to the growth maximizing tax rates is -0.259 (-0.039) when ρ is -0.50 (-0.85).

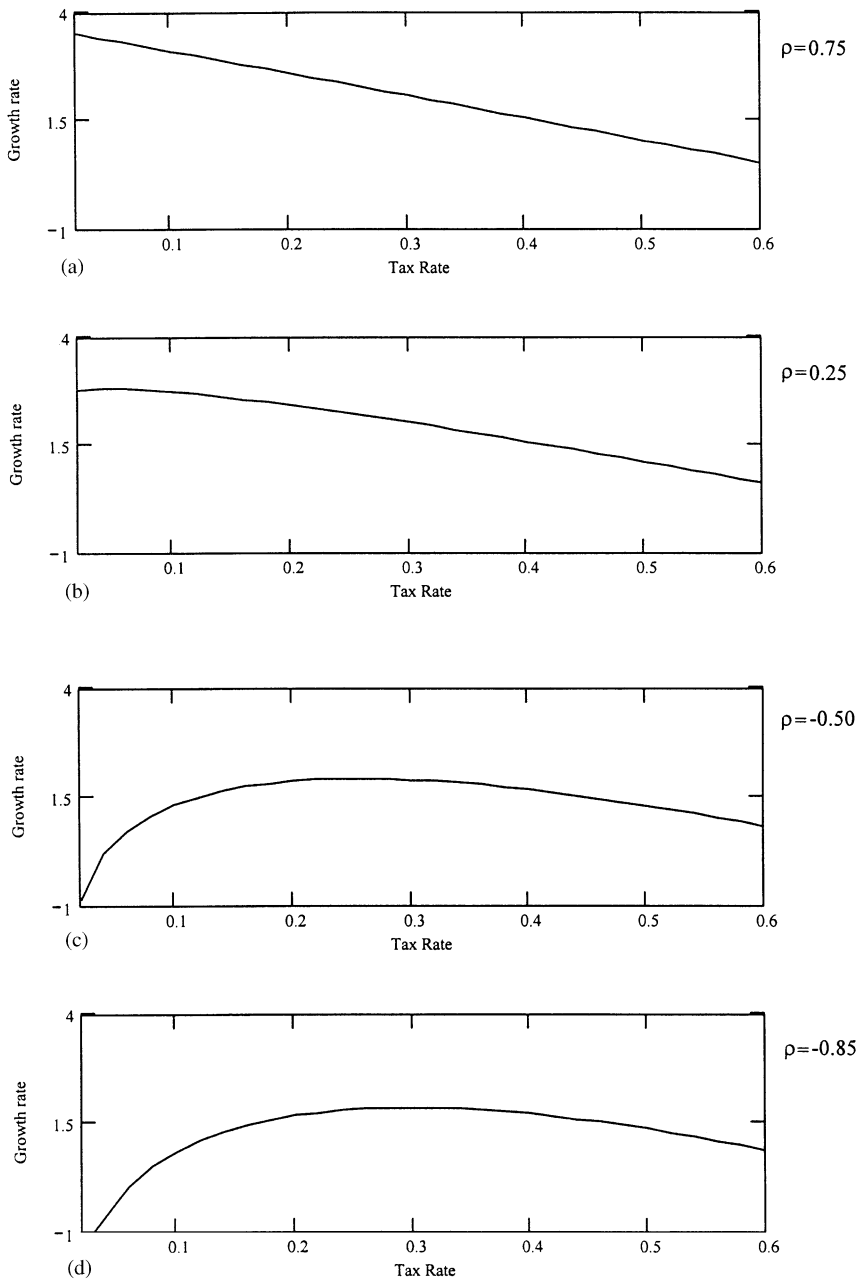


Fig. 5. Growth rate as a function of government's share of output.

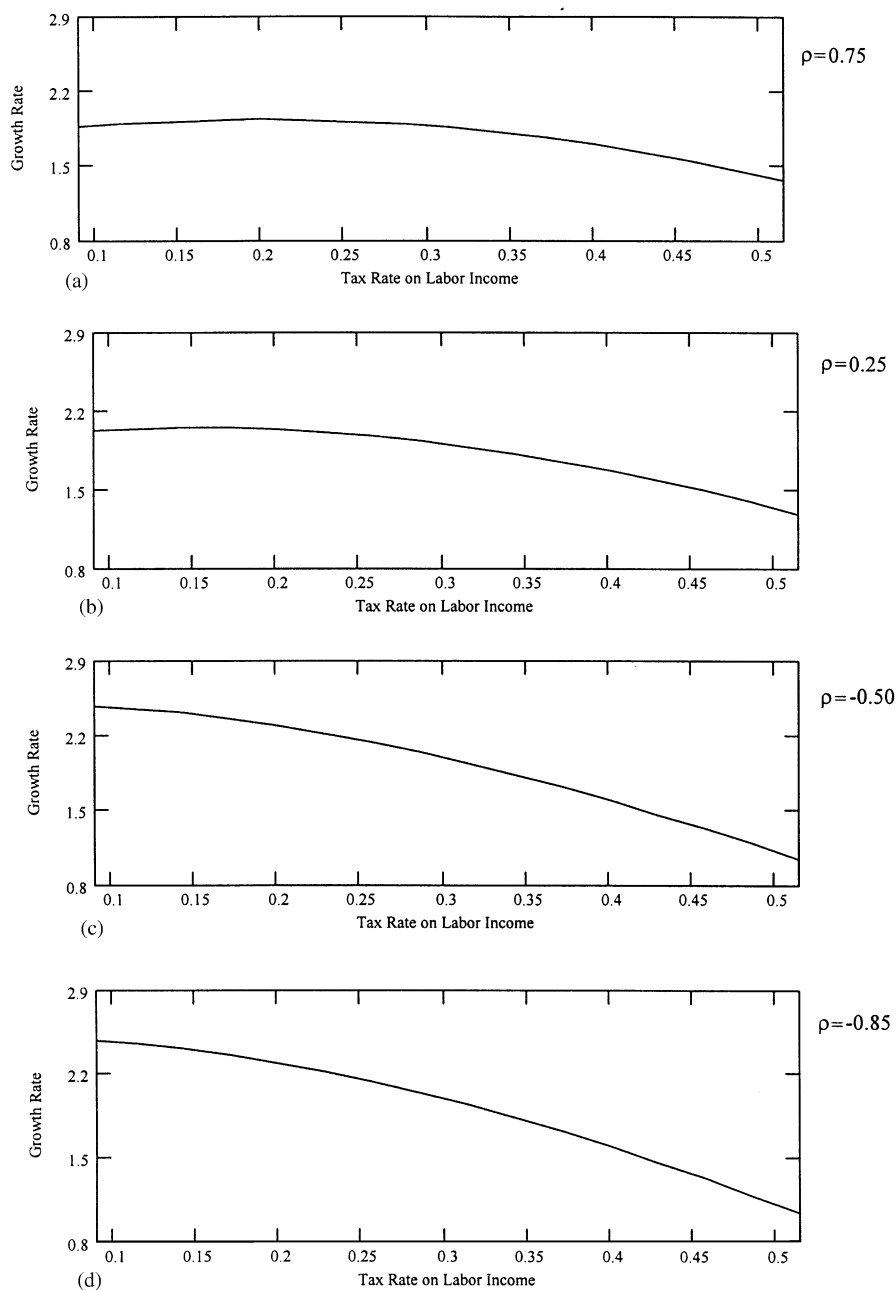


Fig. 6. Growth rate as a function of tax rate on labor income holding government's share constant.

Where the peak occurs depends on the elasticity of substitution between physical capital and public capital. The growth maximizing ratio of public capital to private capital is a decreasing function of ρ . Therefore, as ρ decreases the growth maximizing ratio of public to physical capital increases. The government can achieve this larger ratio by discouraging physical capital accumulation by taxing profits at a higher rate. Accordingly, for lower values of ρ the growth maximizing tax rate on profits (labor income) is higher (lower). Evidently, when $\rho = -0.85$, the growth maximizing tax rate on labor income is zero. When $\rho = -0.50$, the growth rate reaches a maximum (2.49% annually) when labor income is taxed at 5.7% and profits are taxed at 72.9%. In Fig. 6a, the growth rate is maximized when labor income is (profits are) taxed at a rate of 20 (55)%. In this case, the long-run growth rate of output increases to 1.93 (a 7% increase in the growth rate). When $\rho = 0.25$ the long-run growth rate increases by 13.5–2.03% annually when labor income is (profits are) taxed at a rate of 14 (62)%.

Thus, the elasticity of substitution between private inputs and public capital seems to be quantitatively important. When private capital is less substitutable for public capital, the growth maximizing size of the government is large. When private capital and public capital are less substitutable, higher profit taxes will increase the growth rate of the economy.

4. Welfare effects of public policies

The previous sections focused on the growth effects of taxation. The goal of this section is to investigate the welfare effects of public policies. We wish to see if the growth effects shown above have important welfare implications. This is not an exercise in computing the ‘optimal’ fiscal policy.¹⁸ In general, we do not attempt to find combinations of tax rates, rates of infrastructure investment, allocations of publicly provided goods and services, and transfer payments that maximize the household’s lifetime utility. Instead, we report the welfare effects from switching one policy variable at a time and reporting the welfare effects from such a change. Our analysis includes transitional dynamics as well as the welfare effects along the balanced growth path.¹⁹ As in the previous section, we begin with a special case where closed-form solutions can be obtained. This will serve as an appropriate benchmark when we generalize preferences and technology.

¹⁸ A partial list of papers that describe the optimal policies are Jones et al. (1993, 1997), and Cassou and Lansing (1998a, b).

¹⁹ Turnovsky (1992, 1996), among others, shows that the transitional dynamics can be important for determining welfare.

4.1. Welfare when closed-form solutions can be obtained

We begin with the case where technology is Cobb–Douglas ($\rho = 0$), and where labor is inelastically supplied. Without 100% depreciation, it is difficult to assess the welfare effects because of the transitional dynamics. However, if we assume that $\delta_h = \delta_g = \delta_k = 1$, $\sigma = 1$, and identical flat-rate taxes on labor income and profits, then closed-form solutions can be obtained and the welfare effects from public policies are transparent. In this case (14) can be rearranged to yield

$$\gamma = (m_1(\alpha\tau_k + (1 - \alpha)\tau_L))^{\alpha\varphi}(\alpha(1 - \varphi)\beta(1 - \tau_k))^{\alpha(1 - \varphi)}((1 - \alpha)(1 - \tau_L)\beta)^{1 - \alpha}.$$

Consumption is a constant fraction of output and grows at a constant rate; that is

$$c_t = [1 - \alpha(1 - \varphi)\beta(1 - \tau_k) - (1 - \alpha)\beta(1 - \tau_L) - (m_1 + m_2)(\alpha\tau_k + (1 - \alpha)\tau_L)]Y_0\gamma^t.$$

Government services, also, are a constant fraction of output and grow at a constant rate; that is,

$$S_t = m_2(\alpha\tau_k + (1 - \alpha)\tau_L)Y_0\gamma^t.$$

Substituting these expressions into the household's utility function and normalizing time 0 output to unity yields

$$\begin{aligned} U = & \frac{\mu}{(1 - \beta)} \ln(1 - \alpha(1 - \varphi)\beta(1 - \tau_k) - (1 - \alpha)\beta(1 - \tau_L)) \\ & - (m_1 + m_2)(\alpha\tau_k + (1 - \alpha)\tau_L)) + \frac{(1 - \mu)}{1 - \beta} \ln((m_2)(\alpha\tau_k + (1 - \alpha)\tau_L)) \\ & + \frac{\beta}{(1 - \beta)^2} \ln(\gamma). \end{aligned}$$

Intuitively, since all households are the same they would never want to be taxed and then have a portion of tax revenues redistributed back to them in the form of a lump-sum transfers. Therefore, the welfare maximizing public policy will include no transfer payments. This implies $m_1 + m_2 = 1$.

The optimal allocation of tax revenues between public investment and the provision of government services is given by

$$m_1 = \frac{\alpha\varphi\beta}{(1 - \mu)(1 - \beta) + \alpha\varphi\beta}, \quad m_2 = \frac{(1 - \mu)(1 - \beta)}{(1 - \mu)(1 - \beta) + \alpha\varphi\beta}.$$

For most plausible parameters, the percent of revenues allocated to public infrastructure investment is greater than the percent of revenues allocated to the

provision of government services. Thus, the growth effects from public infrastructure investment seem to be more important than the static utility enhancing effects from the provision of public services.

The government's optimal share of output in this case is given by

$$s = [(1 - \mu)(1 - \beta) + \alpha\varphi\beta].$$

Government's share of output is larger than in the models of Barro (1990) and Glomm and Ravikumar (1994) because government services enter into the utility function. Note if $\mu = 1$, then the model is identical to Glomm and Ravikumar the government's share of national output that maximizes welfare is $\alpha\varphi\beta$.

The rest of this section examines the growth and welfare effects as we relax these restrictive assumptions placed on preferences and technology. More specifically, our first experiment investigates the welfare effects from changing the composition of government spending; that is, we increase (decrease) the fraction of revenues allocated to infrastructure investment (m_1) and decrease (increase) the fraction of revenues that gets allocated to utility augmenting government services keeping (m_2) government expenditures and transfers constant. We separate the welfare changes that are due to transitional dynamics and the welfare effects due to changes in the balanced growth path.²⁰

All parameter values are the same as our benchmark parameters. The only exception is we allow the parameter value on government services in the utility function to differ from zero.²¹ In addition, we report the results when φ is 0.15 as well as when φ is 0.30. Table 1 reports the welfare effects for different combinations of m_1 and m_2 . The top row in each cell gives the percentage change in welfare, the second row in each cell is the percent change in welfare due to transitional dynamics, and the last row gives the percentage increase (decrease) in consumption required to leave the individual as well off in the current regime. When φ is 0.30, welfare increases in all cases if a higher percentage of government revenues is allocated to infrastructure investment. Notice, the welfare change from modest changes in infrastructure investment are substantial. Not surprisingly, the greatest welfare gains occur when $\mu = 1$ and $m_1 = 0.54$. In this case, it would take a 22% increase in consumption per period to leave the individual as well off under the old regime. Two reasons account for this increase in welfare: first is the higher (balanced) growth rate. Second, there is an initial increase in the consumption in response to the policy change. Therefore, not

²⁰ The transitional dynamics are calculated by shooting backward from the terminal balanced growth path to the initial state. The corresponding policy functions were then fit with a fifth-order Chebychev polynomial. This procedure is outlined in Judd (1998).

²¹ Since little is known about this value we experiment with three different values $\mu = \{0.85, 0.925, 1\}$.

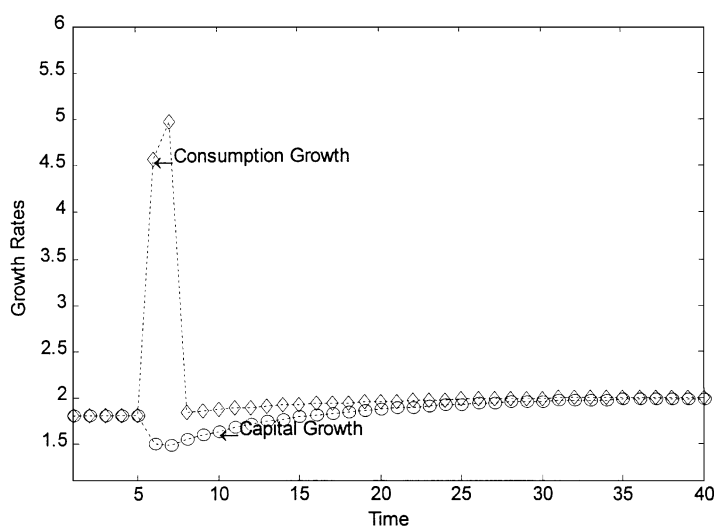


Fig. 7. Growth rates of consumption and capital when infrastructure investment increases.

only do the balanced growth effects raise welfare, the transitional dynamics work to raise welfare as well. In all cases, the lion's share of the welfare effects are due to the transitional dynamics. When $\varphi = 0.15$, the welfare effects are still mostly positive — albeit smaller. The only exceptions occur when households value publicly provided services and only a small fraction of government revenues are allocated to publicly provided services; that is, when $\mu = 0.925$ ($\mu = 0.85$) and $m_2 = 0.10$ ($m_2 = \{0.10, 0.20\}$).

Fig. 7 depicts the transitional dynamics for an increase in infrastructure investment from $m_1 = 0.14$ to $m_1 = 0.24$ and when $\varphi = 0.30$. Consumption growth increases immediately in response to the increase in public infrastructure investment due to the wealth effect of higher lifetime income. The growth rate of the physical capital stock, on the other hand, initially falls. The reason for this seemingly counter-intuitive result is that the increase in the infrastructure investment has little effect on the initial stock of public capital. Therefore, the initial increase in the marginal product of physical and human capital is small. Evidently, the wealth effect (causing consumption to increase) is greater than the effect of higher marginal products on physical and human capital (causing investment to increase). It is only after some period of time that enough public capital accumulates so that growth rate of physical capital increases to the new balanced growth rate.

Table 2 reports the results when we allow for different elasticities of substitution between physical capital and public capital. In all but two cases, welfare is increased by increasing (decreasing) the percent of revenues that are allocated to

infrastructure investment. When public capital and physical capital are more complementary ($\rho = -0.50$) the welfare effects are quite substantial. As one would expect, if public capital and private capital are more substitutable ($\rho = 0.25$) the welfare effects of increasing public investment are smaller. Similar welfare effects were obtained when $\varphi = 0.15$. As in the Cobb–Douglas case, when $\mu = 0.925$ ($\mu = 0.85$) welfare is reduced when infrastructure investment is 0.54 (0.44 and 0.54)% of government revenues.

In Table 3, we illustrate how welfare changes in relation to the size of the government.²² We experiment with tax rates ranging from 0.10 to 0.40. For each tax rate, we compute the change in welfare brought about by the new fiscal policy. As before, we decompose the welfare changes into the fraction that comes from transitional dynamics and the fraction that comes from the new balanced growth path. The tax policy that attains the highest level of welfare is not the tax policy that maximizes the balanced growth rate. When $\mu = 1$ and $\varphi = 0.30$, the percentage change in welfare is higher when the tax rate is reduced to 20% rather than 10% — a value that is closer to the growth maximizing tax rate. The reason for this is because of the importance of the transitional dynamics. The reduction in the tax rates to 10% raises the after-tax marginal product of physical and human capital and leads to large increases in investment. Along the transition path, consumption growth is slowed due to the high returns from investment. This has the effect of reducing the welfare gains. Table 3 also shows that smaller tax cuts result in larger welfare gains when households place a higher value on government services. When $\mu = 1$ and $\varphi = 0.15$, the welfare gains are greatest when the tax rate is reduced to 10%. As in the case with $\varphi = 0.30$, welfare is always (never) increased when tax rates are decreased (increased).

Fig. 8 shows the response of consumption growth and physical capital growth relative to the balanced growth rate along the transition path when taxes are reduced to 25%. As one would expect from the previous discussion the decrease in taxes increases the after-tax marginal product of capital and capital growth is initially high (15% higher than its balanced growth rate). In most other studies in the transition, consumption falls in response to lower taxes. The reason for this is because all tax revenues are given back to the households as lump-sum transfers or are ‘thrown in the ocean’. Here, however, consumption growth starts above its balanced growth rate. There are two reasons for this: First, the wealth effect causes consumption to increase. Second, the reduction in government size implies that government expenditures fall as well. This reduction in resources consumed by the government allows consumption and capital to initially grow at a faster rate than the balanced growth rate.

²² Here we increase the tax rate on labor and capital proportionately.

Table 3
Welfare effects from changes in government size (Cobb–Douglas case)

		$\varphi = 0.30$				$\varphi = 0.15$			
		$\mu = 1$		$\mu = 0.925$		$\mu = 0.85$		$\mu = 1$	
$\tau = 0.10$	% Δ in Welfare (%)	16.45		8.97		3.11		22.93	
	Transition (%)	88		88		88		95	
$\tau = 0.15$	% Δ in Welfare (%)	15.22		10.45		7.48		19.66	
	Transition (%)	84		84		84		89	
$\tau = 0.20$	% Δ in Welfare (%)	12.50		10.39		9.41		15.79	
	Transition (%)	77		79		79		84	
$\tau = 0.25$	% Δ in Welfare (%)	9.11		8.77		9.16		11.19	
	Transition (%)	69		71		70		75	
$\tau = 0.30$	% Δ in Welfare (%)	4.56		5.63		7.15		5.96	
	Transition (%)	51		52		53		57	
$\tau = 0.40$	% Δ in Welfare (%)	–6.95		–3.35		0.66		–7.41	
	Transition (%)	58		58		58		47	
	% Δ in Consumption (%)	19.69		10.70		3.78		29.76	
		17.95		12.67		9.57		24.47	
	% Δ in Consumption (%)	14.29		12.59		12.33		18.75	
		10.02		10.43		11.96		12.61	
	% Δ in Consumption (%)	4.77		6.46		9.12		6.34	
		–6.50		–3.50		0.78		–6.90	
	% Δ in Consumption (%)	5.23		19.30		5.23		15.07	
		6.29		14.20		8.90		12.17	
	% Δ in Consumption (%)	7.94		18.01		7.34		8.90	
		5.84		10.60		4.05		76	
	% Δ in Consumption (%)	1.06		4.66		1.06		5.23	
		–8.37		–6.33		–8.37		–6.33	
	% Δ in Consumption (%)	53		53		53		53	
		–9.02		–6.42		–9.02		–6.42	

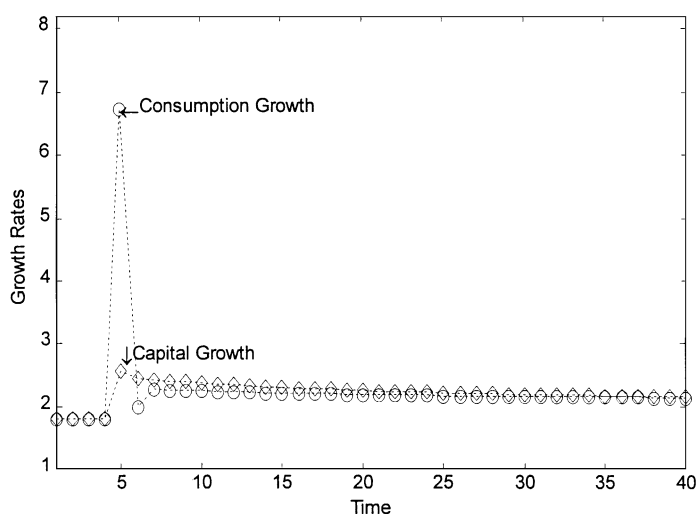


Fig. 8. Consumption and capital growth in response to tax = 0.25.

Table 4 reports the results when we allow for different elasticities of substitution between physical capital and public capital. The welfare effects are largest when public capital and private capital are closer substitutes ($\rho = 0.25$). The reason for this is that when private capital and public capital are closer substitutes it is better to have the private sector accumulate capital than have the public sector accumulate capital through distortionary taxation. Accordingly, a smaller government size reduces the distortions on private decisions and encourages more private capital to be accumulated. Moreover, the welfare effects of reducing government size are enhanced when $\varphi = 0.15$. When public capital and private capital are less substitutable ($\rho = -0.50$), the welfare effects are significantly smaller. In fact, when $\mu = 0.85$ welfare is lowered by reducing the tax rate to 10%. In sum, if public capital and private capital are less (more) substitutable reductions in the size of the government have negligible (sizeable) welfare implications.

5. Conclusion

In a model similar in spirit to Barro (1990), Glomm and Ravikumar (1994), and Turnovsky and Fisher (1995), this paper analyzes the growth and welfare effects of public policies. First, we find growth and welfare effects of increasing infrastructure investment can be substantial. These welfare effects can be substantial even when increasing infrastructure investment comes at the expense of

Table 4
Welfare effects from changes in government size

	$\rho = -0.50$											
	$\rho = 0.25$				$\rho = 0.15$				$\rho = 0.30$			
	$\varphi = 0.30$				$\varphi = 0.15$				$\varphi = 0.15$			
	$\mu = 1$	$\mu = 0.925$	$\mu = 0.85$	$\mu = 0.85$	$\mu = 1$	$\mu = 0.925$	$\mu = 0.85$	$\mu = 0.85$	$\mu = 1$	$\mu = 0.925$	$\mu = 0.85$	$\mu = 0.85$
$\tau = 0.10$												
% Δ in Welfare (%)	21.11	10.92	1.34	8.56	24.98	16.71	8.56	– 9.41	19.92	11.68	3.42	
Transition (%)	91	92	92	94	94	94	94	70	86	85	85	
% Δ in Consumption (%)	26.77	12.46	1.56	11.11	33.06	21.84	11.11	10.09	12.54	14.36	4.17	
$\tau = 0.15$												
% Δ in Welfare (%)	17.89	10.82	4.82	9.78	21.33	15.43	9.78	0.10	14.61	12.82	7.30	
Transition (%)	87	88	88	91	91	91	91	65	82	83	81	
% Δ in Consumption (%)	21.79	13.18	5.99	12.88	27.12	19.87	12.88	0.12	12.11	15.99	9.33	
$\tau = 0.20$												
% Δ in Welfare (%)	14.41	9.31	6.14	9.11	17.06	12.96	9.11	2.68	12.18	11.29	8.18	
Transition (%)	81	83	83	87	86	87	87	61	78	75	75	
% Δ in Consumption (%)	16.83	11.14	7.74	11.89	20.57	16.18	11.89	3.25	13.88	13.82	10.56	
$\tau = 0.25$												
% Δ in Welfare (%)	10.20	7.27	5.07	7.16	12.31	9.54	7.16	3.27	9.34	8.78	6.87	
Transition (%)	73	75	76	78	78	78	78	51	68	66	66	
% Δ in Consumption (%)	11.35	8.50	6.31	9.13	14.04	11.45	9.13	3.99	10.30	10.45	8.74	
$\tau = 0.30$												
% Δ in Welfare (%)	5.73	3.56	2.29	3.99	6.43	5.27	3.99	2.21	5.25	4.68	3.68	
Transition (%)	55	57	57	60	59	60	60	35	49	47	47	
% Δ in Consumption (%)	6.07	4.00	2.76	4.91	6.88	6.03	4.91	2.66	5.54	5.32	4.51	
$\tau = 0.40$												
% Δ in Welfare (%)	– 7.20	– 5.06	– 5.09	– 5.43	– 7.49	– 6.56	– 5.43	– 5.18	– 6.31	– 6.31	– 5.31	
Transition (%)	51	50	51	47	47	47	47	28	60	63	63	
% Δ in Consumption (%)	– 6.72	– 5.20	– 5.68	– 6.04	– 6.98	– 6.64	– 6.04	– 5.77	– 5.93	– 6.40	– 5.91	

reducing the amount of government revenues allocated to public services that augment utility. Second, unlike Stokey and Rebelo (1995), who do not find elasticity of substitution between private inputs quantitatively important, we find elasticity of substitution among *public and private* inputs is important. This elasticity of substitution matters for how revenues are raised and is important for how governments allocate their expenditures. Depending on the elasticity of substitution, the growth and welfare implications of different public policies can be substantial. Thus, in order to implement fiscal policies that raise the growth rate or increase welfare of the individuals more empirical work needs to be done to determine the elasticity of substitution between public and private inputs. Finally, we find the effects of public policy are magnified when labor is elastically supplied.

Appendix

Proposition 1. With m_2 , τ_L , and τ_k held constant, the growth rate is monotonically increasing infrastructure investment.

Proof. Taking the logs of both sides of (14) and differentiating, we obtain

$$\frac{\partial \gamma}{\partial m_1} = \frac{\alpha \varphi}{m_1} \Gamma^{-1}, \quad (\text{A.1})$$

where

$$\Gamma = \left[\frac{\alpha}{\gamma - (1 - \delta_g)} + \frac{\alpha(1 - \varphi)\sigma\gamma^{\sigma-1}}{\gamma^\sigma - \beta(1 - \delta_k)} + \frac{(1 - \alpha)\sigma\gamma^{\sigma-1}}{\gamma^\sigma - \beta(1 - \delta_k)} \right]$$

where (A.1) is positive as long as the gross growth rate of output exceeds unity. $\square$

Proposition 2. For fixed m_1, m_2 , and for $\tau_k = \tau_L = \tau$, the tax rate that maximizes growth is given by

$$\tau = \alpha \varphi.$$

Proof. Taking the logs of both sides of (14) and differentiating, we obtain

$$\frac{\partial \gamma}{\partial \tau} = \left[\frac{\alpha \varphi - \tau}{(1 - \tau)(\tau)} \right] \Gamma^{-1}.$$

If

$$\begin{aligned}\tau < \alpha\varphi &\Rightarrow \frac{\partial\gamma}{\partial\tau} > 0, \\ \tau = \alpha\varphi &\Rightarrow \frac{\partial\gamma}{\partial\tau} = 0, \\ \tau > \alpha\varphi &\Rightarrow \frac{\partial\gamma}{\partial\tau} < 0\end{aligned}\tag{A.2}$$

and the second-order sufficiency condition indicates $\tau = \alpha\varphi$ is a maximum. $\square$

Proposition 3. For fixed m_1, m_2 , and s , the tax rates that maximize the growth rate are:

$$\begin{aligned}\tau_k &= [1 - \varphi)s + \varphi(1 - \alpha)]/(1 - \alpha\varphi), \\ \tau_L &= s - \alpha\varphi.\end{aligned}$$

Proof. This follows from substituting $\tau_L = (s - \alpha\tau_k)/(1 - \alpha)$ into (14) and differentiating yields:

$$\frac{\partial\gamma}{\partial\tau_k} = \frac{\alpha[(1 - \varphi)s + (1 - \alpha)\varphi - (1 - \alpha\varphi)\tau_k]}{(1 - \tau_k)[(1 - \alpha)(1 - \tau_L)]} \Gamma^{-1}.$$

If

$$\tau_k < [(1 - \varphi)s + (1 - \alpha)\varphi]/(1 - \alpha\varphi) \Rightarrow \frac{\partial\gamma}{\partial\tau_k} > 0.$$

If

$$\tau_k = [(1 - \varphi)s + (1 - \alpha)\varphi]/(1 - \alpha\varphi) \Rightarrow \frac{\partial\gamma}{\partial\tau_k} = 0.$$

If

$$\tau_k > [(1 - \varphi)s + (1 - \alpha)\varphi]/(1 - \alpha\varphi) \Rightarrow \frac{\partial\gamma}{\partial\tau_k} < 0$$

and the second-order sufficiency condition indicates:

$$\tau_k = [(1 - \varphi)s + (1 - \alpha)\varphi]/(1 - \alpha\varphi) \text{ is a maximum.}$$

References

- Ai, C., Cassou, S., 1995. A normative analysis of public capital. *Applied Economics* 27, 1201–1209.
Barro, R., 1990. Government spending in a simple model of endogenous growth. *Journal of Political Economy* 98, S103–S125.

- Berndt, E.R., Hansson, B., 1991. Measuring the contribution of public infrastructure capital in Sweden. Manuscript.
- Cassou, S., Lansing, K., 1997. Growth effects of a flat tax. Manuscript.
- Cassou, S., Lansing, K., 1998a. Optimal fiscal policy, public capital, and the productivity slowdown. *Journal of Economic Dynamics and Control* 22, 911–935.
- Cassou, S., Lansing, K., 1998b. Optimal tax reform and public-sector investment in human capital. Manuscript.
- Christiano, L., 1991. Modeling the liquidity effect of a money shock. *Federal Reserve Bank of Minneapolis Quarterly Review*, Winter.
- Frenkel, J., Razin, A., 1996. *Fiscal Policies and Growth in the World Economy*. MIT Press, Cambridge, MA.
- Garcia-Mila, T., McGuire, T., 1992. The contribution of publicly provided inputs to states' economies. *Journal of Regional Science* 22, 229–241.
- Glomm, G., Ravikumar, B., 1994. Public investment in infrastructure in a simple growth model. *Journal of Economic Dynamics and Control* 18, 1173–1187.
- Holtz-Eakin, D., 1994. Public sector capital and the productivity puzzle. *The Review of Economic Statistics* 76, 12–21.
- Jones, L., Manuelli, R., Rossi, P., 1993. Optimal taxation in models of endogenous growth. *Journal of Political Economy* 101, 485–517.
- Jones, L., Manuelli, R., Rossi, P., 1997. On the optimal taxation of capital income. *Journal of Economic Theory* 73, 93–117.
- Judd, K., 1998. *Numerical Methods in Economics*. MIT Press, Cambridge, MA.
- King, R., Plosser, C., Rebelo, S., 1988. Production, growth, and business cycles I: the basic neoclassical model. *Journal of Monetary Economics* 21, 195–232.
- King, R., Rebelo, S., 1990. Public policy and economic growth: developing neoclassical implications. *Journal of Political Economy* 98, S127–S150.
- Lansing, K., 1998. Optimal fiscal policy in a business cycle model with public capital. *Canadian Journal of Economics* 31, 337–364.
- Lucas Jr., R.E., 1990. Supply-side economics: an analytic review. *Oxford Economic Papers* 42, 293–316.
- Lynde, C., Richmond, J., 1993. Public capital and total factor productivity. *International Economic Review* 34, 401–414.
- Nadiri, M., Mamuneas, T., 1994. The effects of public infrastructure and R&D capital on the cost structure and performance of U.S. manufacturing industries. *The Review of Economics and Statistics* 76, 22–37.
- Pecorino, P., 1993. Tax structure and growth in a model with human capital. *Journal of Public Economics* 52, 251–271.
- Saint-Paul, G., 1992. Fiscal policy in an endogenous growth model. *Quarterly Journal of Economics* 107, 1243–1259.
- Stokey, N., Rebelo, S., 1995. Growth effects of flat-rate taxes. *Journal of Political Economy* 103, 519–550.
- Turnovsky, S., 1992. Alternative forms of government expenditure financing: a comparative welfare analysis. *Economica* 59, 235–252.
- Turnovsky, S., 1996. Optimal tax, debt, and expenditure policies in a growing economy. *Journal of Public Economics* 60, 21–44.
- Turnovsky, S., Fisher, W., 1995. The composition of government expenditure and its consequence for macroeconomic performance. *Journal of Economic Dynamics and Control* 19, 747–786.